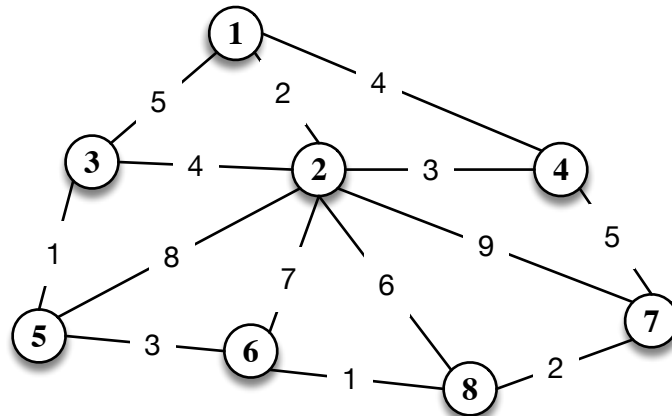


First part

Score: 1) 4 pts, 2) 5 pts, 3) 4 pts, 4) 4 pts.

1. Consider the minimum spanning tree problem in the figure below. Determine the optimal solution showing the intermediate steps.

For arcs $(2,7)$ e $(2,3)$, determine for which values of the cost coefficients the solution remains optimal. Justify your answer.



2. Consider the following linear programming problem:

$$\begin{aligned} \max \quad & -3x_1 \\ & -x_1 + x_2 \leq 4 \\ & -2x_1 - x_2 \leq 2 \\ & -2x_1 + x_2 \leq 4 \end{aligned}$$

Consider the two subsets of inequality indices: $B_1 = \{1, 2\}$ and $B_2 = \{2, 3\}$. For each of them discuss the primal and the dual feasibility conditions.

3. Consider the following Integer Linear Programming problem:

$$\begin{aligned} \max \quad & -2x_1 + x_2 \\ & -2x_1 + 4x_2 \leq 7 \\ & 6x_1 + 8x_2 \leq 27 \\ & x_1, x_2 \geq 0, \text{ integer} \end{aligned}$$

Add the slack variables (x_3 and x_4) to the first and the second constraints, and consider the optimal base $B = \{2, 4\}$ of the LP relaxation. Generate the Gomory cuts associated with the fractional components of the solution. Justify your answer and give the geometric representation of the feasible region and of the cut(s).

4. Consider the pair of dual problems $\min cx : Ax \leq b, x \in R^n$ and $\max yb, yA = c, y \leq 0, y \in R^m$. Prove that, given two feasible solutions $\bar{x}$ and $\bar{y}$, we have that $c\bar{x} \geq \bar{y}b$.

min	max
variables	constraints
constraints	variables
cost vector c	known term vector b
known term vector b	cost vector c
$A_i x \geq b_i$	$y_i \geq 0$
$A_i x \leq b_i$	$y_i \leq 0$
$A_i x = b_i$	$y_i \geq 0$
$x_i \geq 0$	$yA^i \leq c_i$
$x_i \leq 0$	$yA^i \geq c_i$
$x_i \geq 0$	$yA^i = c_i$

Fig. 8: Primal/Dual correspondence table